

HOLGUIN, CubaWMO: **782623**Lat: **20.786N**Lon: **76.315W**Elev: **110**StdP: **100.01**Time Zone: **-5.00 (CUB)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.0	16.8	10.9	8.2	22.6	12.0	8.8	23.3	10.2	25.6	9.4	25.9	1.1	50	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	34.8	23.8	34.1	23.8	33.2	23.8	26.3	31.0	26.0	30.7	25.6	30.3	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.5	28.1	24.8	20.1	27.9	24.2	19.3	27.3	82.1	31.4	80.3	30.7	78.8	30.5	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.3	7.5	12.1	35.9	1.9	0.6	10.7	36.4	9.5	36.7	8.5	37.1	7.0	37.5
			11.0	27.5	1.8	1.3	9.7	28.5	8.7	29.2	7.7	29.9	6.5	30.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	23.5	24.2	24.9	26.0	27.0	28.0	28.5	28.4	27.9	26.8	25.0	24.2
	DBStd	2.29	1.96	1.89	1.97	1.53	1.29	1.14	0.88	0.96	0.88	1.23	1.66	1.88
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	5918	418	398	462	479	528	541	573	571	536	522	449	441
	CDD18.3	2877	159	165	204	229	269	291	314	313	286	264	199	183
	CDH23.3	25014	1021	1242	1629	1962	2378	2863	3273	3198	2646	2192	1400	1211
(14) Wind	WSAvg	3.4	3.6	3.8	3.9	3.9	3.5	3.2	3.6	3.4	2.7	2.8	3.2	3.4
	PrecAvg	1097	38	36	41	64	142	133	106	127	146	140	79	45
	PrecMax	1392	108	91	114	112	275	247	173	190	270	315	214	179
	PrecMin	737	11	9	14	31	74	69	55	80	76	51	11	14
	PrecStd	134	20	22	21	22	52	42	28	28	38	63	48	30
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	33.0	34.0	34.1	34.9	35.1	35.2	35.2	34.9	33.8	32.5	31.9
		MCWB	20.7	20.6	21.2	21.7	23.3	23.9	24.1	24.1	23.7	23.8	23.5	22.1
	2%	DB	30.6	31.9	32.8	33.0	33.8	34.0	34.2	34.2	34.0	32.9	31.2	30.8
		MCWB	21.0	20.4	21.4	22.2	23.0	24.2	24.1	24.3	24.1	24.2	23.6	22.5
	5%	DB	29.7	30.8	31.8	32.0	32.8	33.1	33.8	33.9	33.1	32.0	30.1	29.9
		MCWB	21.2	20.8	21.3	21.9	23.1	24.2	24.1	24.3	24.4	24.3	23.4	22.5
	10%	DB	28.2	29.8	30.2	31.0	31.8	32.2	32.9	33.0	32.2	31.1	29.1	28.9
		MCWB	21.1	21.1	21.1	21.7	23.1	24.3	24.2	24.4	24.5	24.4	22.9	22.2
	0.4%	WB	23.6	23.5	24.9	24.8	26.2	26.7	26.5	26.7	26.6	26.2	25.6	24.6
		MCDB	27.8	28.4	30.2	30.3	31.4	31.5	31.7	31.7	31.3	30.2	28.8	28.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	22.8	22.8	23.3	23.9	25.4	26.0	25.9	26.1	26.1	25.7	24.8	23.9
		MCDB	26.8	27.3	28.2	28.7	30.6	30.9	31.1	31.0	30.7	29.8	28.6	28.0
	5%	WB	22.4	22.3	22.6	23.3	24.7	25.4	25.4	25.7	25.7	25.2	24.3	23.5
		MCDB	26.3	26.8	27.5	28.3	29.6	30.2	30.4	30.4	30.2	29.5	28.1	27.4
	10%	WB	22.0	21.8	22.1	22.8	24.2	25.0	25.0	25.2	25.2	24.8	23.9	23.0
		MCDB	25.8	26.3	26.9	27.7	28.7	29.5	29.8	29.8	29.7	29.0	27.4	26.6
	5% DB	MDBR	9.8	10.7	11.0	10.8	10.1	9.6	10.0	10.3	10.2	9.1	8.8	9.1
		MCDBR	11.4	12.3	12.5	11.9	11.3	10.4	10.9	11.1	10.9	10.1	9.7	10.2
		MCWBR	4.0	3.8	3.6	3.3	3.4	3.4	3.2	3.4	3.6	3.6	3.7	3.8
		MCWBR	9.4	10.6	11.0	10.8	9.7	9.8	10.1	10.2	10.1	9.2	8.4	9.0
(40) Clear-Sky Solar Irradiance	taub	0.366	0.372	0.380	0.419	0.440	0.476	0.483	0.455	0.423	0.404	0.384	0.382	0.382
	taud	2.519	2.495	2.475	2.366	2.346	2.237	2.203	2.311	2.421	2.477	2.491	2.488	2.488
	Ebn at Noon	909	923	926	888	859	822	817	845	872	879	882	878	878
	Edh at Noon	98	106	112	126	128	141	146	132	116	106	100	98	98
(44) All-Sky Solar Radiation	RadAvg	4.18	5.03	5.68	6.08	5.92	6.03	6.32	6.11	5.62	4.77	4.24	3.97	3.97
	RadStd	0.29	0.33	0.30	0.35	0.35	0.31	0.28	0.16	0.31	0.34	0.29	0.27	0.27

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	+1.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page